

Bounded Semivariation Forces Analyticity for Locally Convex Semigroups

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Status

Candidate full solution, likely valid. The proof answers Problem 4.17 of Kruse–Schwenninger [2], subject to checking the exact formulation of Dembart’s analyticity criterion [1, Theorem 1, p.154].

Source Problem

The source is Karsten Kruse and Felix L. Schwenninger, *Continuous maximal regularity in locally convex spaces*, arXiv:2408.11437 [2]. Problem 4.17 appears on PDF page 25.

In the case that X is a Banach space, it also follows from Corollary 4.6 that $(T(t))_{t \geq 0}$ is analytic (holomorphic) if it is a strongly continuous semigroup on X such that $(T(t))_{t \in [0, r]}$ is of bounded semivariation for some $r > 0$ (see [81, Lemma 3.3, p. 429]). Is this still true in our more general setting where one might use one of the three equivalent assertions in [23, Theorem 1, p. 154] to define analyticity of a semigroup on a sequentially complete complex Hausdorff locally convex space?

4.17. Problem. Let $r > 0$, X be a sequentially complete Hausdorff locally convex space over the field $\mathbb{C}$ and $(T(t))_{t \geq 0}$ a strongly continuous locally equicontinuous semigroup on X . Is $(T(t))_{t \geq 0}$ an analytic semigroup if $(T(t))_{t \in [0, r]}$ is of bounded semivariation?

In transcription, the problem asks:

Let $r > 0$, X be a sequentially complete Hausdorff locally convex space over the field $\mathbb{C}$ and $(T(t))_{t \geq 0}$ a strongly continuous locally equicontinuous semigroup on X . Is $(T(t))_{t \geq 0}$ an analytic semigroup if $(T(t))_{t \in [0, r]}$ is of bounded semivariation?

Definitions and Source Facts Used

Let A be the generator of $(T(t))_{t \geq 0}$ in the sense used in [2]. A family $(T(t))_{t \in [0, r]}$ is of bounded semivariation if for every continuous seminorm q on X there is a continuous seminorm p on X such that

$$SV_{q,p}^{[0,r]}(T) = \sup_d \sup_{p(x_i) \leq 1} q \left(\sum_i (T(d_i) - T(d_{i-1}))x_i \right) < \infty,$$

where d ranges over finite partitions of $[0, r]$.

We use three facts from the source paper. First, bounded semivariation on one nontrivial interval implies bounded semivariation on every compact interval [2, Remark 4.1(a)]. Second, bounded semivariation implies $T(t)X \subset D(A)$ for every $t > 0$ [2, Corollary 4.6]. Third, under the same hypothesis the mild solution of the homogeneous Cauchy problem is a classical solution on $(0, r]$ [2, Corollary 4.16]; in particular, for each $x \in X$ the orbit $t \mapsto T(t)x$ is differentiable for $t > 0$ and

$$\frac{d}{dt}T(t)x = AT(t)x.$$

Finally, we use Dembart's analyticity criterion in the form explicitly pointed to by the source: for a strongly continuous locally equicontinuous semigroup on a sequentially complete complex Hausdorff locally convex space, analyticity is equivalent to positive-time domain inclusion together with local equicontinuity of the family

$$\{tAT(t) : 0 < t \leq R\} \subset \mathcal{L}(X)$$

for every $R > 0$ [1, Theorem 1, p.154].

Proof Intuition

The only missing estimate is uniform in t control of $tAT(t)$. Bounded semivariation controls arbitrary sums of increments $T(d_i) - T(d_{i-1})$ applied to vectors whose seminorm is bounded. For a fixed t , choose the uniform partition of $[0, t]$ and feed into the semivariation sum the vectors $T(t - d_i)x$. The semigroup identity makes every summand equal to the same backward difference $T(t)x - T(t - t/n)x$. The whole sum is therefore $n(T(t) - T(t - t/n))x$, which converges to $tAT(t)x$. Local equicontinuity of the original semigroup controls the auxiliary vectors $T(t - d_i)x$, so the semivariation bound passes to the limit.

Theorem and Proof

Theorem 1. *Let X be a sequentially complete complex Hausdorff locally convex space and let $(T(t))_{t \geq 0}$ be a strongly continuous locally equicontinuous semigroup on X . If $(T(t))_{t \in [0, r]}$ is of bounded semivariation for some $r > 0$, then $(T(t))_{t \geq 0}$ is analytic in Dembart's sense.*

Proof. By [2, Remark 4.1(a)], the family $(T(t))_{t \in [0, R]}$ is of bounded semivariation for every $R > 0$. Fix such an R . We prove that

$$\mathcal{F}_R := \{tAT(t) : 0 < t \leq R\}$$

is an equicontinuous subset of $\mathcal{L}(X)$.

Let q be a continuous seminorm on X . By bounded semivariation on $[0, R]$ there is a continuous seminorm p on X and a constant

$$V := SV_{q,p}^{[0,R]}(T) < \infty$$

such that, for every partition $0 = d_0 < \dots < d_n \leq R$ and every finite choice of vectors $y_i \in X$,

$$q \left(\sum_{i=1}^n (T(d_i) - T(d_{i-1}))y_i \right) \leq V \max_{1 \leq i \leq n} p(y_i).$$

The displayed estimate is just the homogeneous extension of the defining semivariation bound.

Since the semigroup is locally equicontinuous, there are a continuous seminorm p_0 on X and a constant $C \geq 0$ such that

$$p(T(s)x) \leq Cp_0(x), \quad 0 \leq s \leq R, \quad x \in X.$$

Now fix $t \in (0, R]$, $x \in X$, and $n \in \mathbb{N}$. Put $d_i = it/n$ and $y_i = T(t - d_i)x$. Then d is a partition of $[0, t] \subset [0, R]$ and $p(y_i) \leq Cp_0(x)$. Therefore

$$q \left(\sum_{i=1}^n (T(d_i) - T(d_{i-1}))T(t - d_i)x \right) \leq VCp_0(x).$$

By the semigroup identity,

$$(T(d_i) - T(d_{i-1}))T(t - d_i)x = T(t)x - T(t - t/n)x$$

for every i . Thus

$$q(n(T(t) - T(t - t/n))x) \leq VCp_0(x).$$

By [2, Corollary 4.16], $s \mapsto T(s)x$ is differentiable for $s > 0$ with derivative $AT(s)x$. Hence

$$n(T(t) - T(t - t/n))x = t \frac{T(t)x - T(t - t/n)x}{t/n} \longrightarrow tAT(t)x$$

in X . Since q is continuous, passing to the limit gives

$$q(tAT(t)x) \leq VCp_0(x), \quad 0 < t \leq R, \quad x \in X.$$

This proves equicontinuity of $\mathcal{F}_R$.

Also, $T(t)X \subset D(A)$ for every $t > 0$ by [2, Corollary 4.6]. Dembart's criterion [1, Theorem 1, p.154] now yields that $(T(t))_{t \geq 0}$ is analytic. $\square$

Verification Notes

The proof is formal and uses no computational search. The key estimate is insensitive to the choice of the fundamental seminorm system. The case $\max_i p(y_i) = 0$ in the homogeneous semivariation extension is handled by scaling: the defining bound applies to $\lambda^{-1}y_i$ for every $\lambda > 0$, and then $\lambda \downarrow 0$.

The main point for human review is the final invocation of Dembart. The source paper explicitly proposes Dembart's Theorem 1 as the appropriate definition of analyticity in this setting; this packet uses the standard formulation with positive-time domain inclusion and local equicontinuity of $\{tAT(t)\}$.

Novelty Check

The run indexes were searched for arXiv:2408.11437, continuous maximal regularity, locally convex spaces, bounded semivariation, analytic semigroup, and Dembart. No previous run packet for Problem 4.17 was found. A bounded web search on 2026-06-29 for exact phrases around “bounded semivariation”, “analytic semigroup”, “locally convex”, and “Dembart” did not reveal a separate later answer.

References

References

- [1] B. Dembart. On the theory of semigroups of operators on locally convex spaces. *Journal of Functional Analysis*, 16(2):123–160, 1974.
- [2] K. Kruse and F. L. Schwenninger. Continuous maximal regularity in locally convex spaces. [arXiv:2408.11437](#), 2024.